UNIVERSITY OF SCIENCE AND TECHNOLOGY
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Lab Activity 3

IT325 Machine Learning | Training and Comparing Five Models

Type	Laboratory Activity
Topic Focus	Model comparison for text classification
Time Allotment	1 laboratory meeting
PIT Connection	Students produce a formal comparison of candidate models for their dataset
Mode	Individual hands on work

Learning Outcomes

1. Transform text data into numerical features using a suitable vectorization method.
2. Train at least five classification models using the same dataset split.
3. Evaluate models using the same metrics and identify strong candidates for tuning.

Required Workflow

1. Prepare features and target variables.
2. Split the dataset into training and testing sets.
3. Train five models using the same prepared data.
4. Evaluate each model using accuracy, precision, recall, F1 score, and confusion matrix.
5. Summarize results in one comparison table.

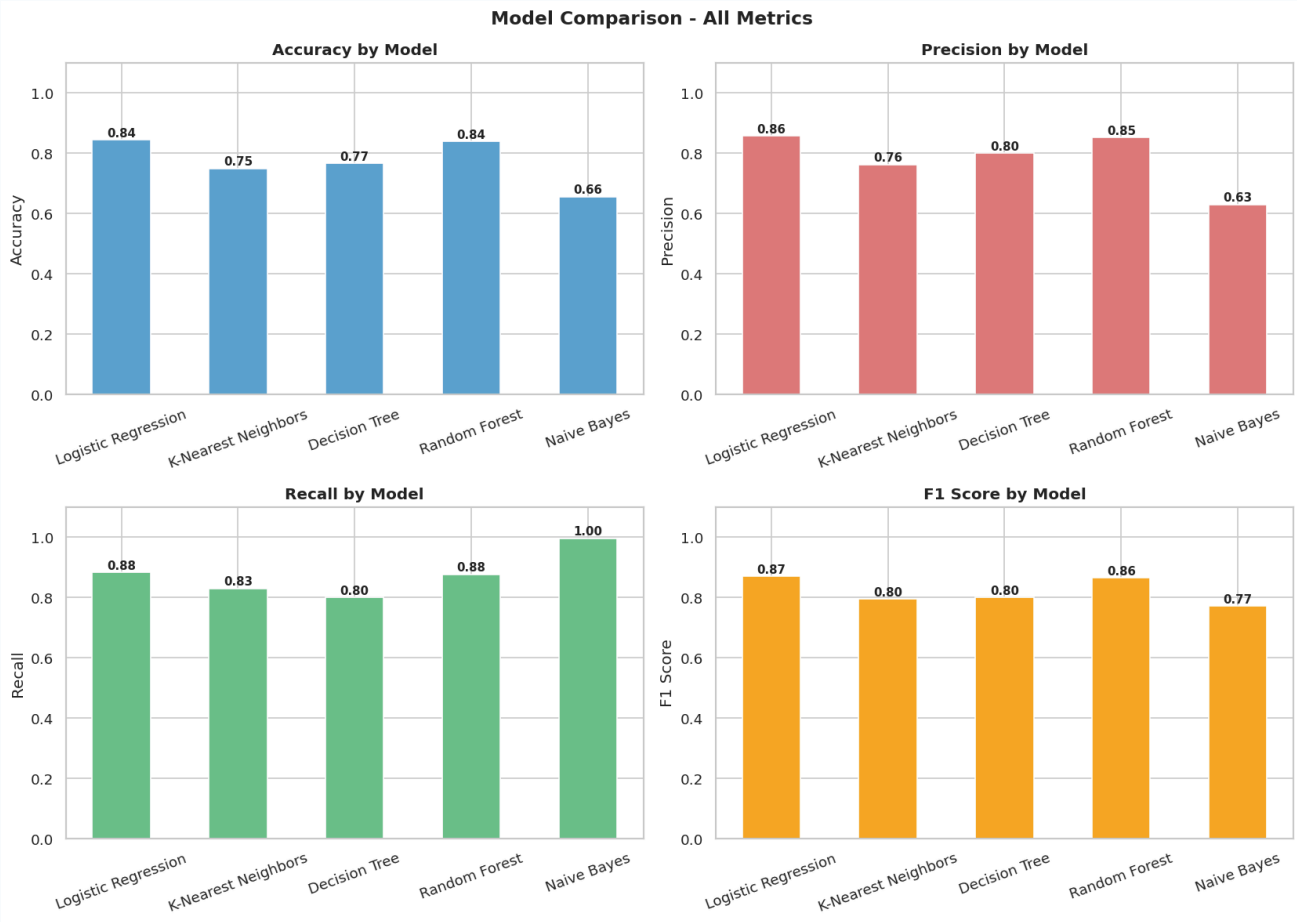
Note: Use exactly the same dataset split and evaluation approach for all models so that the comparison remains fair and meaningful.

Choose any 3 of the Models Discussed

1. Linear Regression
2. Logistic Regression
3. K Nearest Neighbors
4. Decision Tree
5. Random Forest
6. Naive Bayes

Required Write Up

1. Present one comparison table for the five models.



2. State which two models will move to the tuning stage.

I would move Logistic Regression and Random Forest to the tuning stage.

3. Write one paragraph explaining the basis for your selection.

I selected Logistic Regression and Random Forest as the top two models based on their overall performance across all evaluation metrics. Logistic Regression achieved the highest accuracy at 84.48% and the highest F1 score at 0.8694, making it the most balanced and reliable model overall.

Random Forest came in second with an accuracy of 83.87% and an F1 score of 0.8643, which is very close to Logistic Regression and still strong across precision and recall. Both models outperformed K-Nearest Neighbors, Decision Tree, and Naive Bayes in terms of balanced performance.

While Naive Bayes had the highest recall at 0.9960, its accuracy of only 65.57% means it incorrectly classifies too many non-depressed students as depressed, making it unreliable for practical use. Logistic Regression and Random Forest strike the best balance between catching actual depressed students and avoiding false alarms, which is why I chose them for the tuning stage.

Link of the Colab Notebook:	Gonzales - Lecture_Activity_3_Model_Comparison.ipynb

Simple Scoring Guide

Use the checklist below to evaluate the submitted work for this activity.

Criteria	Description	Points
Model Implementation	Five models are trained correctly.	10
Evaluation	Required metrics and comparison table are complete.	10
Selection Rationale	Top models are chosen with sound justification.	5